

Implementing a health-system specialty pharmacist-led pharmacogenomic protocol for DPYD genetic testing to optimize capecitabine safety

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Background

- DPYD gene variants can lead to a deficiency in dihydropyrimidine dehydrogenase, an enzyme involved in the breakdown of fluoropyrimidine-based chemotherapy agents (e.g., capecitabine), and increase the risk of severe adverse events 16-fold.¹
- Preemptive pharmacogenomic testing can mitigate these risks and is recommended to identify high-risk patients prior to therapy initiation.²
- Adoption is limited in the US due to challenges such as cost and reimbursement, inadequate healthcare training, and conflicting recommendations in clinical guidelines.³
- Health-system specialty pharmacy (HSSP) teams can play a key role in ensuring medication safety and optimal outcomes by confirming that pharmacogenomic testing was completed, when appropriate, prior to initiating medication.

Objective

To describe an HSSP’s role in a multidisciplinary protocol for preemptive pharmacogenomic testing of DPYD metabolizer status prior to capecitabine initiation.

Description

Quality Improvement Initiative

An ongoing quality improvement initiative was launched in 2024 at a health system whose specialty pharmacy services are managed by CPS Solutions, LLC (CPS). This initiative adopted a multidisciplinary approach to ensure that proactive pharmacogenomic testing for DPYD metabolizer status was performed prior to initiating capecitabine.

Evaluation Measures

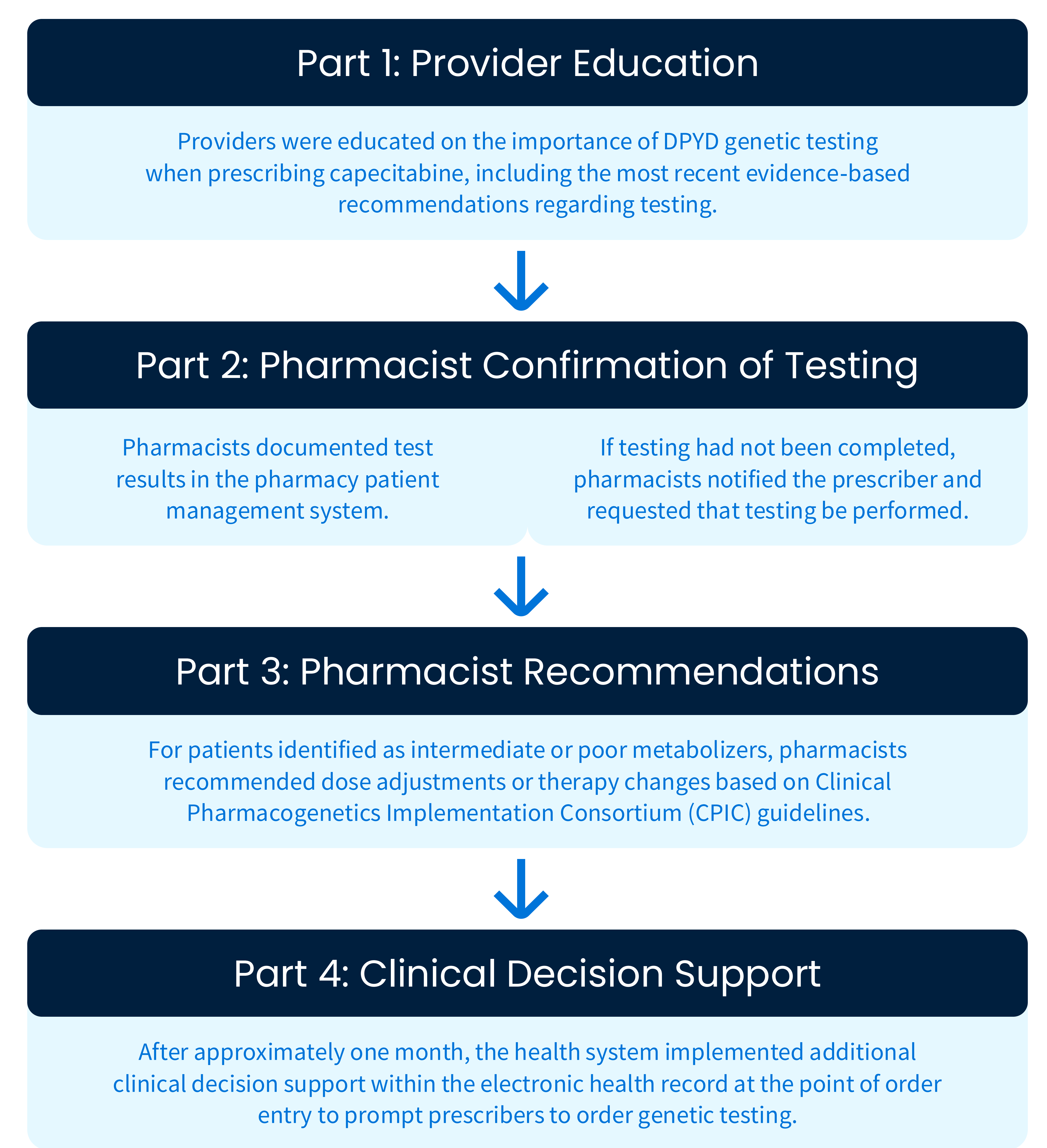
- Number of patients evaluated for DPYD genetic testing prior to capecitabine initiation
- Number of patients at increased risk of toxicity
- Recommendations made by pharmacists
- Outcomes of pharmacist recommendations

Data Analysis

- Descriptive statistics were used to present all measures. Data through February 2025 was included for analysis.

Description

Figure 1: Description of Quality Improvement Initiative



Evaluation

To date, 30 patients were evaluated by the HSSP team for DPYD genetic testing prior to initiating capecitabine. Pharmacists recommended dose adjustments for high-risk patients to mitigate adverse effects, while also providing patients with supportive medications and education on the increased risks.

- 10% Three (10%) patients were identified as intermediate metabolizers and at a significantly increased risk of severe toxicity.
- 100% All recommendations were communicated to the oncologists, documented in the patient management system, and accepted by the providers.

Discussion & Conclusion

- HSSP teams are valuable in the implementation of pharmacogenomic protocols into clinical workflows due to their expertise in medication management and precision medicine.
- HSSP pharmacists can play a crucial role in optimizing chemotherapy regimens, reducing toxicities, and improving patient outcomes.

Next Steps

- The next steps in this initiative will focus on evaluating how this protocol can be adapted and scaled to other health systems who manage patients on capecitabine.
- Future initiatives will consider other areas where pharmacogenomic testing could be a valuable part of patient management protocols within HSSPs.

References

¹ De Mattia E, Silvestri M, Polesel J, et al. Rare genetic variant burden in DPYD predicts severe fluoropyrimidine-related toxicity risk. *BiomedPharmacother*. 2022;154:113644. doi:10.1016/j.biopha.2022.113644

² Chenchula S, Atal S, Uppugunduri CRS. A review of real-world evidence on preemptive pharmacogenomic testing for preventing adverse drug reactions: a reality for future health care. *Pharmacogenomics J* . 2024;24(2):9. doi:10.1038/s41397-024-00326-1

³ Virelli CR, Mohiuddin AG, Kennedy JL. Barriers to clinical adoption of pharmacogenomic testing in psychiatry: a critical analysis. *Transl Psychiatry*. 2021 Oct 6;11(1):509. doi: 10.1038/s41398-021-01600-7. PMID: 34615849; PMCID: PMC8492820.